

Bayesian multimodeling: Distributions, expectation, likelihood

2025

Random variable

Given:

- A set of elementary events Ω
- A sigma-algebra $\mathfrak{F}$ over Ω
- A system $\mathfrak{B}(\mathbb{R})$ of Borel sets over $\mathbb{R}$

A real-valued function $w(\omega) : \Omega \rightarrow \mathbb{R}$ is called random variable, whenever for each $B \in \mathfrak{B}(\mathbb{R})$:

$$\{\omega : w(\omega) \in B\} \in \mathfrak{F}.$$

Random variable characteristics

A discrete random variable w takes a countable set of values $A = \{a_1, a_2, \dots\}$ with probabilities $p_1, p_2, \dots$, $\sum_i p_i = 1$.

$f(a_i) = p(w = a_i) = p_i$ is a **probability function**.

A continuous random variable is set using **cumulative distribution function (CDF)**:

$$F_w(t) = p(w \leq t)$$

or **probability density function (PDF)**:

$$f(w) : \int_a^b f(w) dw = p(a \leq w \leq b).$$

Maximum likelihood

$$X \sim f(x, w),$$

$$L(X, w) = \prod_{x \in X} f(x, w),$$

$$\hat{w} \equiv \arg \max_w L(X, w).$$

We can also use a logarithmic form:

$$\log L(X, w) = \sum_{x \in X} \log f(x, w),$$

$$\hat{w} \equiv \arg \max_w \log L(X, w).$$

Maximum likelihood variations

Score function:

$$S(w) \equiv \frac{\partial}{\partial w} \log L(w)$$

Maximum likelihood estimator is a solution of the score equation:

$$S(w) = 0$$

Fisher information:

$$I(w) \equiv -\frac{\partial^2}{\partial w^2} \log L(w)$$

MLE dispersion:

$$\mathbb{D}\hat{\theta} \approx I^{-1}(\hat{w})$$

MLE properties

- Consistency:

$$\hat{w}_n \rightarrow^P w$$

- Asymptotic normality: $n \rightarrow \infty$

$$\hat{w} \sim \mathcal{N}(w, I^{-1}(w))$$

- Efficiency: MLE's dispersion is minimal over all the unbiased estimations
- Invariance: $g(\hat{w})$ is MLE over $g(w)$

Likelihood maximization

Likelihood maximization is a KL divergence minimization:

$$\max_w L(X, w) \iff \min KL(p^*(X) | p(X|w)).$$

Proof sketch

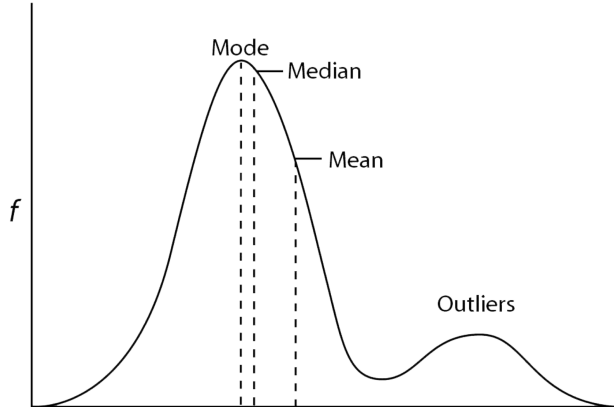
$$\begin{aligned} KL(p^*(X) | p(X|w)) &= E_{x \sim p^*(X)} \log \left(\frac{p^*(X)}{p(X|w)} \right) = \\ &= \text{Const} - E_{x \sim p^*(X)} \log p(X|w) \approx^{\text{Law of large numbers}} \\ &\approx \text{Const} - L(X, w). \end{aligned}$$

Central tendency

Empirical mean is an arithmetic mean for the given data.

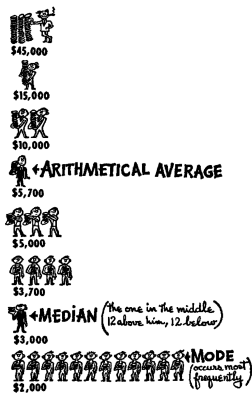
Empirical median is a central element in the variational series.

Empirical mode is the most popular element in the data



Central tendency

(Huff, 1954):



Median

- α -quantile:

$$w_{\alpha}: \quad p(w \leq w_{\alpha}) \geq \alpha, \quad p(w \geq w_{\alpha}) \geq 1 - \alpha$$

or, equivalently:

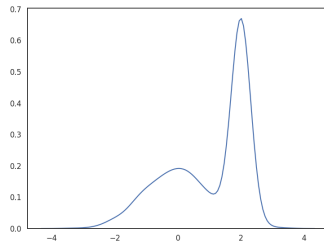
$$w_{\alpha} = F^{-1}(\alpha) = \inf\{w: F(w) \geq \alpha\}$$

- **median** is a 0.5-quantile, the central value of the distribution:

Mode

A mode is a maximum of the CDF or PDF:

$$\text{mode}(w) = \arg \max_w f(w)$$



Distribution can have multiple modes:

$$w \sim \alpha_1 \mathcal{N}(\mu_1, \sigma_1^2) + \alpha_2 \mathcal{N}(\mu_2, \sigma_2^2).$$

Expectation

Average value of the random variable w :

$$Ew = \int w dF(w).$$

- can be undetermined;
- linear;
- does not depend on the 0-measure values;
- Law of large numbers:

$$\bar{w}_n \xrightarrow{n \rightarrow \infty} Ew;$$

- Central Limit Theorem:

$$\sqrt{n} \frac{\bar{w}_n - Ew}{\sqrt{Dw}} \rightarrow \mathcal{N}(0, 1).$$

Expectation: convergence

Often we need to investigate the convergence of the random variable, which is constructed as a composition of functions:

$$w \rightarrow^{d/p/a.s.} w^*,$$

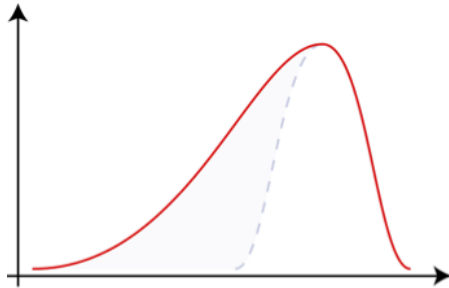
$$Ef(w) \rightarrow Ef(w^*)?$$

- f is continuous and limited, then $Ef(w) \rightarrow^d Ef(w^*)$.
- f is continuous almost everywhere, then $f(w) \rightarrow^{d/p/n.h.} f(w^*)$ (Mann–Wald theorem).
- A.s. convergence: Lebesgue theorem (swap limit and expectation).

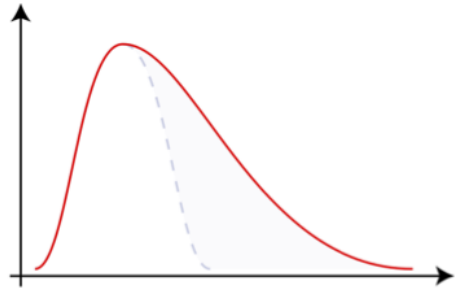
Other moments

- Second moment $Dw = E(w - Ew)^2$: dispersion.
- Third moment $\frac{E(w - Ew)^3}{Dw^{3/2}}$: asymmetry coefficient.
- Forth moment $\frac{E(w - Ew)^4}{Dw^2} - 3$: Excess coefficient, pointedness of the PDF.

Other moments

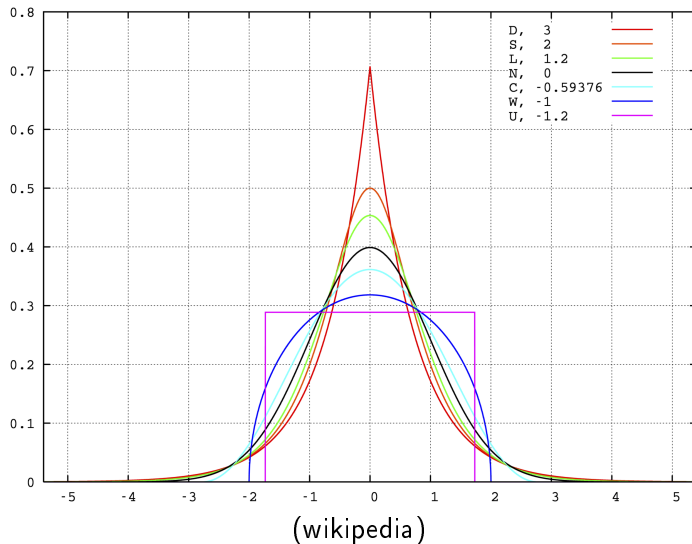


Negative Skew

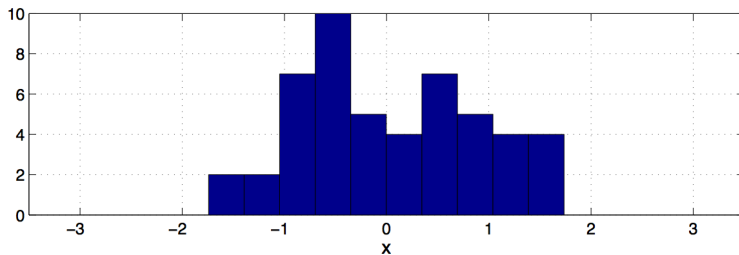
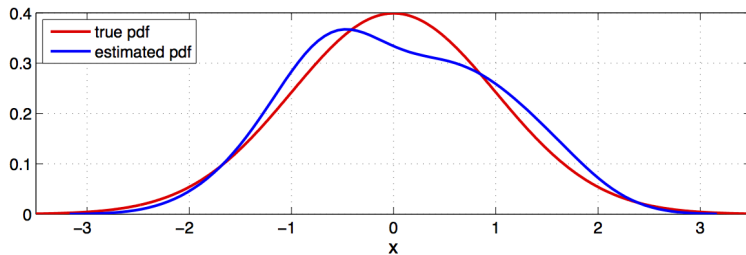


Positive Skew

Other moments



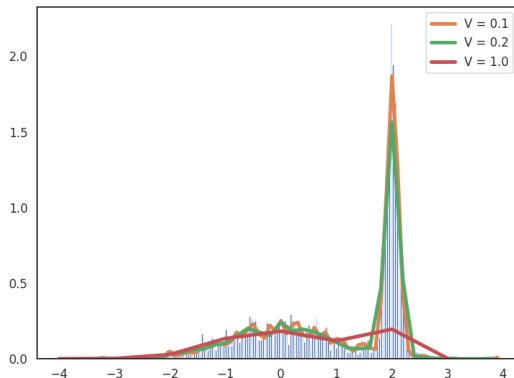
PDF estimation



KDE

Estimation using kernel function K :

$$p(x) = \frac{1}{N} \sum_{x_i \in X} \frac{1}{h} K\left(\frac{x - x_i}{h}\right)$$

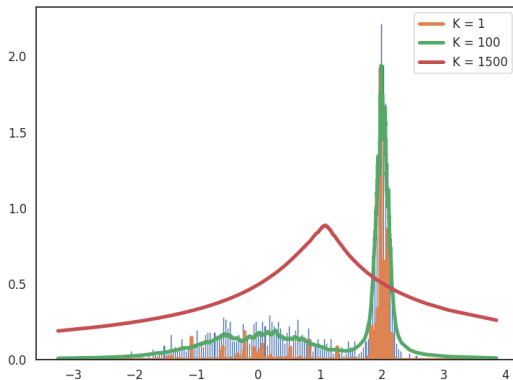


KDE

KNN can be considered as a generalization of KDE:

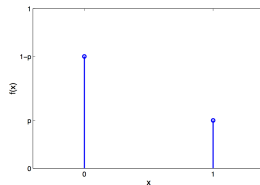
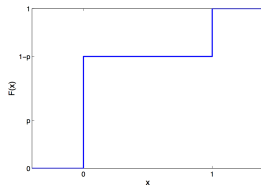
$$p(x) = \frac{1}{N} \sum_{x_i \in X} \frac{1}{h} K\left(\frac{x - x_i}{h_i}\right),$$

$$h_i = \text{dist}(x_i, \text{neighbour}(x_i, K))$$



Bernoulli distribution

$w \in \{0, 1\} \sim \text{Ber}(p)$, $p \in (0, 1)$

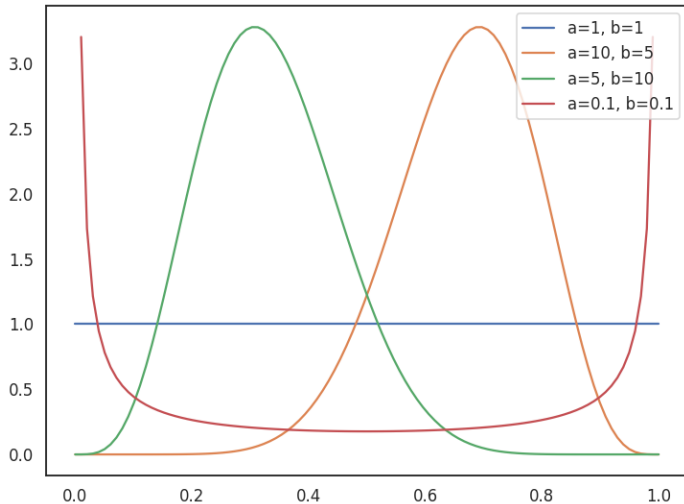


$$F(w) = \begin{cases} 0, & w < 0, \\ 1-p, & 0 \leq w < 1, \\ 1, & w \geq 1. \end{cases}$$
$$f(w) = \begin{cases} 1-p, & w = 0, \\ p, & w = 1. \end{cases}$$

- example: coin flipping

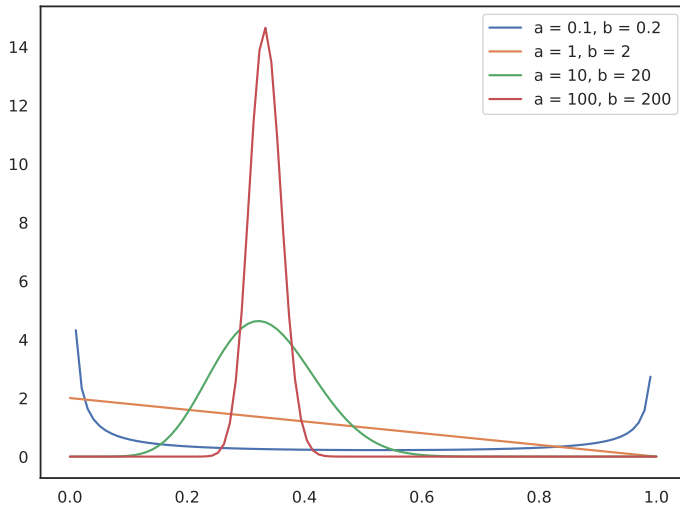
Beta-distribution

$$p(w) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} w^{a-1} (1-w)^{b-1}$$



Beta-distribution

$$p(w) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} w^{a-1} (1-w)^{b-1}$$



Beta-distribution

distribution

- corresponds to the *prior* beliefs about Bernoulli distribution
- interpretation: “effective number of events $w = 1, w = 0$ ”
- With $n \rightarrow \infty$ converges to δ -distribution with PDF concentration at MLE for Bernoulli.

Multinomial distribution

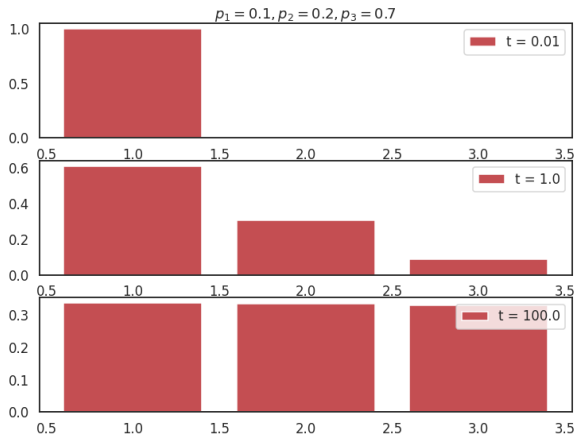
- Generalization of Bernoulli distribution for a larger number of events:

$$p(w = w_i | p_1, \dots, p_n) = p_i.$$

- Can be parametrized using softmax:

$$\text{softmax}(\log p, t) = \frac{\exp(-\log p \cdot t^{-1})}{\sum_{i=1}^n \exp(-\log p_i \cdot t^{-1})}$$

Multinomial distribution



Dirichlet distribution

$$p(w_1, \dots, w_K, \alpha) = \frac{1}{B(\alpha)} \prod_{i=1}^K w_i^{\alpha_i - 1}$$

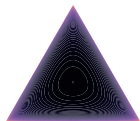
- Generalization of Beta-distribution for a larger number of events
- Support: $K - 1$ -simplex:

$$w_i \geq 0, \sum w_i = 1.$$

- Interpretation: a probability of each of K mutually exclusive events equals to w_i given that each event was observed $\alpha_i - 1$ times
- Can be parametrized using the following expression:

$$\alpha = \bar{\alpha} \cdot t, ||\alpha|| = 1.$$

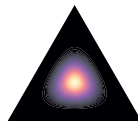
Dirichlet distribution



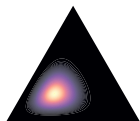
$$\bar{\alpha} = [1, 1, 1], t = 0.9$$



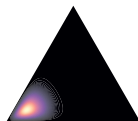
$$t = 1.0$$



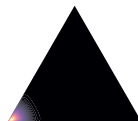
$$t = 10.0$$



$$\bar{\alpha} = [0.5, 0.25, 0.25], t = 30.0$$



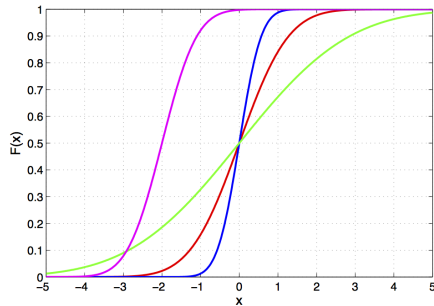
$$[0.75, 0.125, 0.125]$$



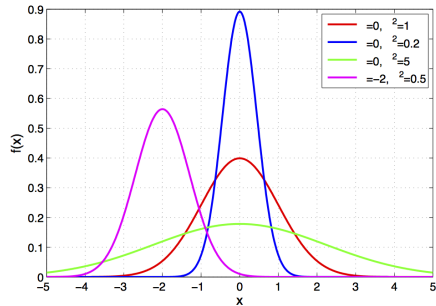
$$[0.9, 0.05, 0.05]$$

Normal distribution

$$w \in \mathbb{R} \sim \mathcal{N}(\mu, \sigma^2), \sigma^2 > 0$$



$$F(w) = \Phi\left(\frac{w - \mu}{\sigma}\right)$$
$$f(w) = \frac{1}{\sigma} \phi\left(\frac{w - \mu}{\sigma}\right)$$



$$\Phi(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^w e^{-\frac{t^2}{2}} dt$$
$$\phi(w) = \frac{1}{\sqrt{2\pi}} e^{-\frac{w^2}{2}}$$

Normal distribution

- limit of the sum of the weakly dependent random variables.
- $Ew = \text{median}(w) = \text{mode}(w) = \mu$, $Dw = \sigma^2$, the higher moments equal to zero
- if $w_1, \dots, w_n$ are independent, $w_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, then $\forall a_1, \dots, a_n$

$$\sum_{i=1}^n a_i w_i \sim \mathcal{N}\left(\sum_{i=1}^n a_i \mu_i, \sum_{i=1}^n a_i^2 \sigma_i^2\right)$$

- Normal distribution has the maximal differential entropy among all the continuous distributions with limited dispersion.
- example: estimation error

Student distribution

- $Ew = 0$ при $\nu > 1$, $\text{median}(w) = \text{mode}(w) = 0$ always
- let $Z \sim N(0, 1)$ and $V \sim \chi^2_\nu$ independent, then

$$T = \frac{Z}{\sqrt{V/\nu}} \sim St(\nu)$$

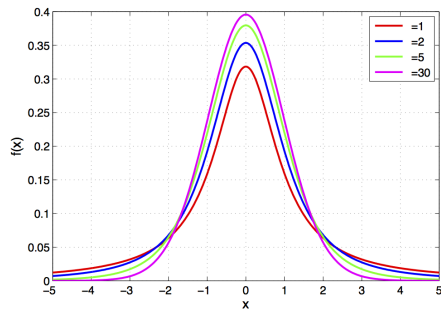
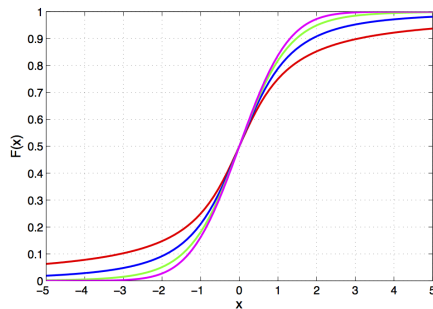
- if $w \sim St(\nu)$, then

$$Y = \lim_{\nu \rightarrow \infty} w \sim \mathcal{N}(0, 1)$$

- can be met during empirical mean estimation

Student distribution

$X \in \mathbb{R} \sim St(\nu), \nu > 0$



$$F(x) = \frac{1}{2} + x \Gamma\left(\frac{\nu+1}{2}\right)$$

$$f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

Periodical distribution

?

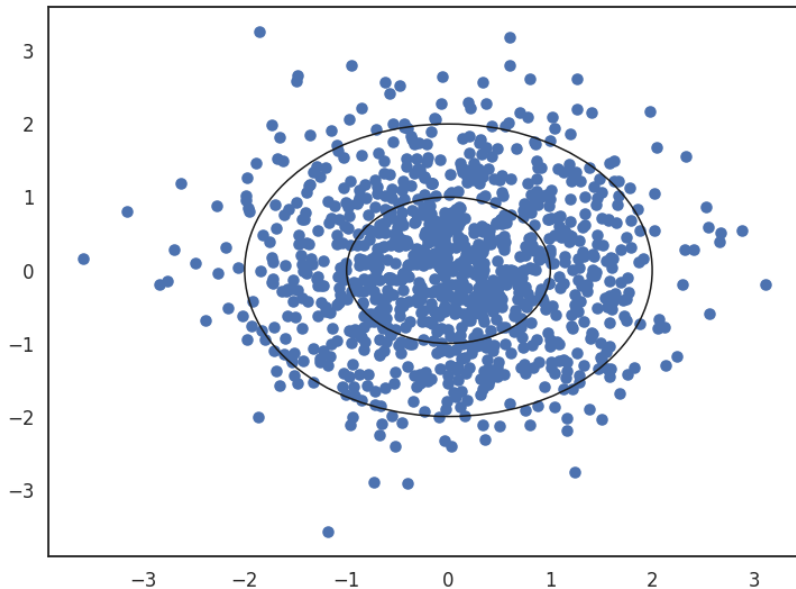
Periodical distribution

Naive approach: use a non-periodical distribution (let's say, Normal?)

Periodical distribution

- What if our dataset contains two objects on the circle: (1 degree and 359 degree)?
- $E_w = 180$, $D = 179$

Periodical distribution



Periodical distribution

- von Mises distribution:

$$p = \frac{\exp(k \cos(x - \mu))}{2\pi I_0(k)}, \quad I_0 \text{ is a Bessel function..}$$

- Distribution idea:

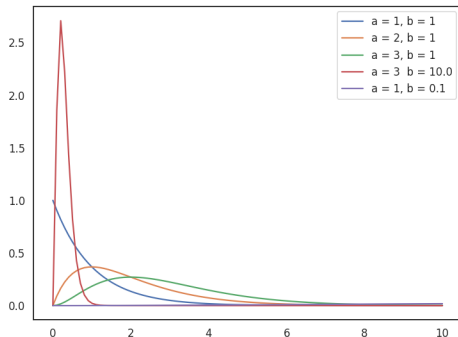
- ▶ $[w_1, w_2] \sim \mathcal{N}(\mu, \sigma^2)$
- ▶ Parameterize using polar coordinate system, limit the radius:

$$w_1 = r \cos \phi, w_2 = r \sin \phi, r = \text{Const.}$$

- See also Anton Bishuk's 2021 task 2 about Kent distribution.

Gamma distribution

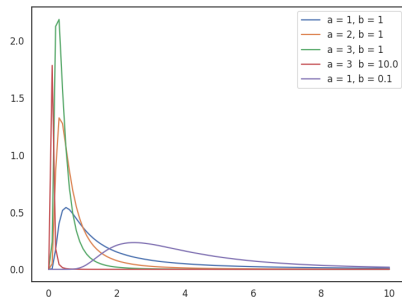
$$p(w, \alpha, \beta) = \frac{\beta^\alpha w^{\alpha-1} e^{-\beta w}}{\Gamma(\alpha)}$$



- Can be used for prior distribution modeling, for the values inverse to standard deviation.

Inverse Gamma-distribution

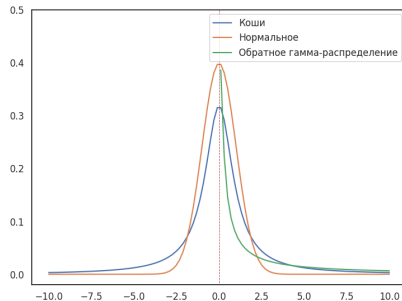
$$p(w, \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} w^{-\alpha-1} \exp(-\beta \cdot w^{-1}).$$



- $w \sim \Gamma(\alpha, \beta) \rightarrow w^{-1} \sim \text{Inv} - \Gamma(\alpha, \beta^{-1})$
- Can be used for prior distribution modeling, for the standard deviation for example.

Cauchy distribution

$$p(w, w_0, \gamma) = \frac{\gamma}{\pi ((w - w_0)^2 + \gamma^2)}.$$



- Alternative to Normal distribution and Inverse Gamma-distribution
- Heavy tails
- The moments are undetermined

References

- Ширяев А. Н. Вероятность-1. – МЦНМО, 2007. – С. 552-552.
- Bishop C. M. Pattern recognition //Machine learning. – 2006. – Т. 128. – №. 9.
- KL minimization: <https://wiseodd.github.io/techblog/2017/01/26/kl-mle/>

Talks

- 2 talks must be presented during the semester
- For each talk the student also provides tex files and 2 questions for Google form
- One talk must be also represented as a blog-post
 - ▶ Target audience: 2 or 3rd year student of our department
 - ▶ Language: English
 - ▶ Timing: up to 10 minutes to read
 - ▶ Focus on illustration of the approach (don't 100 supplementary experiments from appendix)
 - ▶ The resulting blog post will be published in IS website (or on the student's website if they want)
 - ▶ No need to provide blog post ASAP: it can be done later, **BUT**...
 - ▶ ...There will be a cross-review, so keep track of time
- the topics for the next talk are on the course page (talks.md)

Project overview

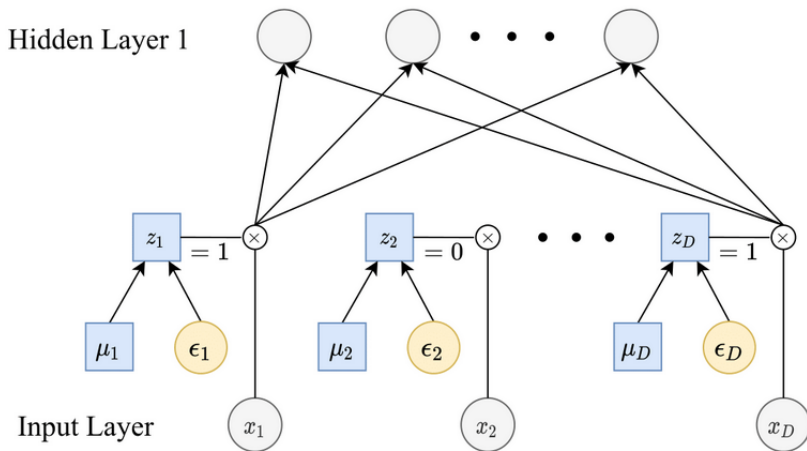
Activities

Each team must assign roles for all teammates. Each activity is evaluated independently, thus number of activities per each teammate must be $\approx$ equal.

- Project planning
- Benchmarks
- Algorithm implementation (**1 activity per teammate**)
- Project wrapping
- Tests writing
- Documentation writing
- Cross-review (**1 activity per teammate**)
- Final demo
- Blog post
- Tech. report

All the activities must be finished according to the project schedule (see course page). For the deadline violation the score will be multiplied by 0.66 for each week.

Stochastic gating



Yamada et al.

Project: Stochastic gating

- Feature selection with L2 regularization and straight-through estimation see here for example , see also basic paper for ST-estimator
- Gumbel-softmax for gating
- Original stochastic gating
- Correlated features in stochastic gating

The team must decide the structure of the project: are the modules they propose are just layer classes (like dropout layers) or they propose some "solvers"? What's more preferable for reproducibility and further usage?

Model selection: coherent Bayesian inference

First level: find optimal parameters:

$$w = \arg \max \frac{p(\mathcal{D}|w)p(w|h)}{p(\mathcal{D}|h)},$$

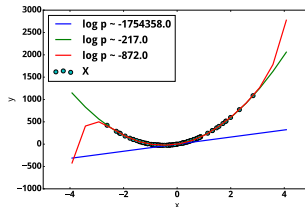
Second level: find optimal model:

Evidence:

$$p(\mathcal{D}|h) = \int_w p(\mathcal{D}|w)p(w|h)dw.$$



Model selection scheme



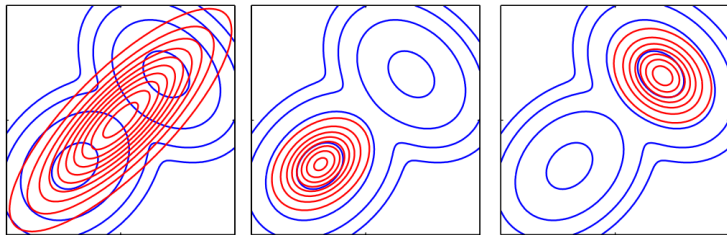
Polynomial regression example

Evidence lower bound, ELBO

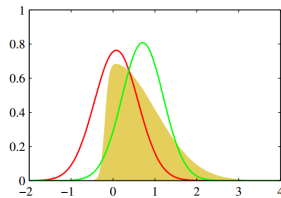
Evidence lower bound is a method of approximation of intractable distribution $p(w|\mathcal{D}, h)$ with a distribution $q(w) \in \mathcal{Q}$.

Evidence lower bound estimation often reduces to optimization problem

$$\begin{aligned} \log p(\mathcal{D}|h) &\geq \\ &\geq \text{KL}(q(w)||p(w|\mathcal{D})) = - \int_w q(w) \log \frac{p(w|\mathcal{D})}{q(w)} dw = \mathbb{E}_w \log p(\mathcal{D}|w) - \text{KL}(q(w)||p(w|h)). \end{aligned}$$



Variational inference vs. expectation propagation (Bishop)



Laplace Approximation vs

Variational inference

Project: Bayesian ensembling¹

Each algorithm of evidence approximation proposes different strategies to sample from posterior.

- A baseline ELBO: proposed by Graves in 2011. Must be implemented with local reparameterization trick , hyperparameter optimization and pruning.
- Alternative method: scalable Laplace approximation
- Renyi divergence: a generalization of the ELBO which can potentially plugged into the algorithms 1 and 2 and make models more extensible
- Bayes by backpropagation: an alternative to ELBO.

¹The details will be discussed during classes on Bayesian inference and variational inference, see previous years

Dirichlet distribution

$$p(w_1, \dots, w_K, \alpha) = \frac{1}{B(\alpha)} \prod_{i=1}^K w_i^{\alpha_i - 1}$$

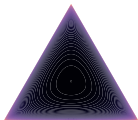
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- Support: $K - 1$ -simplex:

$$w_i \geq 0, \sum w_i = 1.$$

- Interpretation: a probability of each of K mutually exclusive events equals to w_i given that each event was observed $\alpha_i - 1$ times
- Can be parametrized using the following expression:

$$\alpha = \bar{\alpha} \cdot t, ||\alpha|| = 1.$$

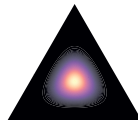
Dirichlet distribution



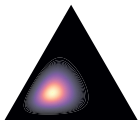
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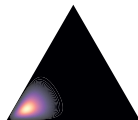
$$t = 1.0$$



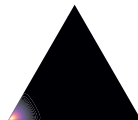
$$t = 10.0$$



$$\bar{\alpha} = [0.5, 0.25, 0.25], t = 30.0$$



$$[0.75, 0.125, 0.125]$$



$$[0.9, 0.05, 0.05]$$

Project: Discrete variables relaxation

In this project the authors need to implement different alternatives to it. **It's a continuation of the previous year project**

- Generalized Gumbel-softmax
- Surrogate-based
- REBAR
- Straight-tought Gumbel-Softmax

The main focus here is to make a good demo/baselines. This is essential here.

Reparametrization trick

Reparamterization idea:

$$\varepsilon = S_{\theta}(w), \quad w = S_{\theta}^{-1}(\varepsilon).$$

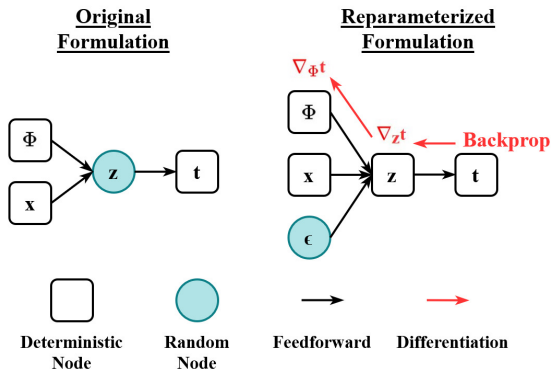
Then:

$$\nabla_{\theta} E_q f(w) = E_q \nabla_{\theta} f(S_{\theta}^{-1}(\varepsilon)).$$

Example:

$$w \sim \mathcal{N}(\mu, \sigma^2) \rightarrow S(w) = \frac{w - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

Challenge: calculation of S^{-1} is an expensive operation.



Source: wikipedia

Project: Low-variance gradient estimation for arbitrary distributions²

The problem reparametrization trick is that it's available only for the limited number of distributions. The goal of this project is to wrap multiple methods describing a solution for this issue into a library performing for (nearly-)arbitrary distributions.

- Rejection-sampling variational inference
- Implicit reparametrization trick
- Transport equation-based reparametrization trick

²The details will be discussed during classes on variational inference, see previous years

Minimum description length principle

$$\text{MDL}(f, \mathcal{D}) = L(f) + L(\mathcal{D}|f),$$

where f is a model, $\mathcal{D}$ is a dataset, L is a description length in bits.

$$\text{MDL}(f, \mathcal{D}) \sim L(f) + L(\mathbf{w}^*|f) + L(\mathcal{D}|\mathbf{w}^*, f),$$

$\mathbf{w}^*$ — optimal parameters.

$\mathbf{f}_1$	$L(\mathbf{f}_1)$	$L(\mathbf{w}_1^* \mathbf{f}_1)$	$L(\mathcal{D} \mathbf{w}_1^*, \mathbf{f}_1)$
$\mathbf{f}_2$	$L(\mathbf{f}_2)$	$L(\mathbf{w}_2^* \mathbf{f}_2)$	$L(\mathcal{D} \mathbf{w}_2^*, \mathbf{f}_2)$
$\mathbf{f}_3$	$L(\mathbf{f}_3)$	$L(\mathbf{w}_3^* \mathbf{f}_3)$	$L(\mathcal{D} \mathbf{w}_3^*, \mathbf{f}_3)$

Project: Neural compression

Information theory and Bayesian models are deeply tied. Here we propose to implement multiple variational and neural compression methods for lossless (and nearly-lossless) compression.

- Var. inference-based
- Gradient flows-based
- Transformer-based
- Lossy compression

Your classifier is secretly an energy based model and you should treat it like one ³

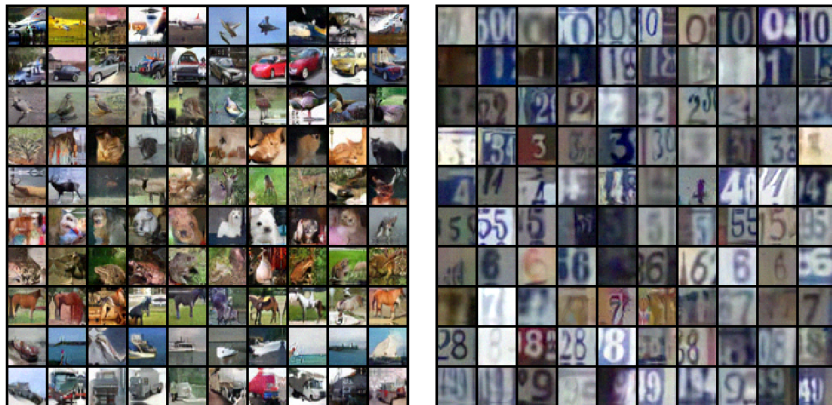


Figure 7: Class-conditional Samples. Left to right: CIFAR10, SVHN.

³The topic will be discussed further. See also the talk of Maria Kovaleva, 2023

Project: Secret energy-based models

The project proposes to consider multiple nearly-standard architectures and reveal their sampling properties: generalized (non-variational) autoencoders, classification neural networks and GAN's discriminator are proven to be energy-based models (EBM) that can be potentially used for generative tasks (sampling, out-of-domain detection, density estimation).

- Denoising autoencoder
- Classifier neural model
- GAN's discriminator

Hometask

- ① The table for project assignment (at yandex disk) will be available tomorrow.
- ② You should write you name and acitvites/roles.
- ③ Kind reminder that after each classes we will have a questionnaire with very simple and trivial questions.